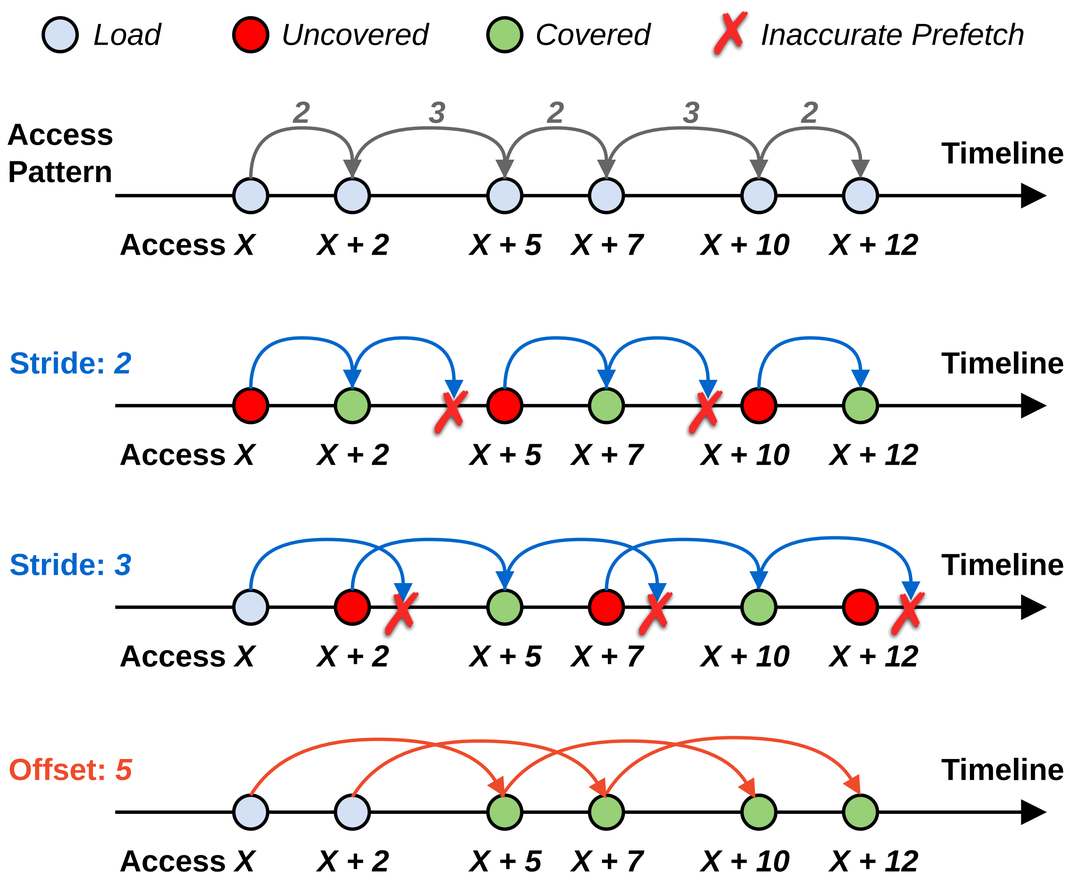
Caerus: Prefetching by Thinking Globally and Acting Locally

Jacky Wong [1] Jacob Alexandrou [1] Yinting Huang [2] Luke Panayi [1] Lev Mukhanov [3] Sam Ainsworth [3] Paul H J Kelly [1]

[1] Imperial College London [2] University of Cambridge [3] IMEC

What is Offset Data Prefetching?

Offset Prefetching is a form of spatial prefetching. Compared to stride prefetching that targets the next access, offset prefetching targets an eventual access.

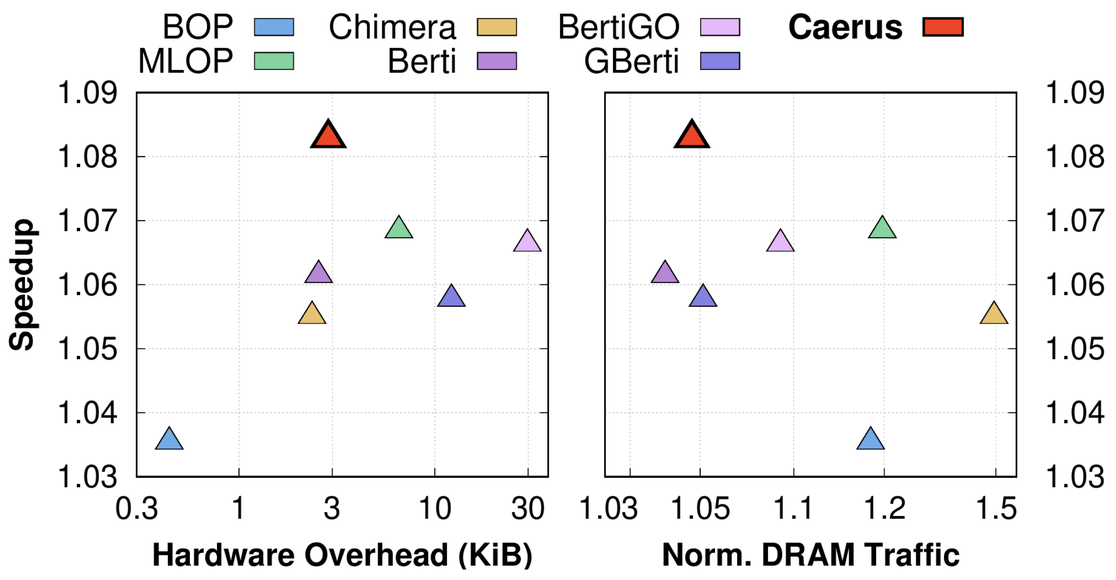


Research Questions

How do we effectively **control** our prefetchers to avoid unecessary contention? (Problem 3 →)

And with this control in mind, how do we completely **redesign our prefetchers** to achieve the maximum performance? (Problems 1, 2 →)

Evaluation

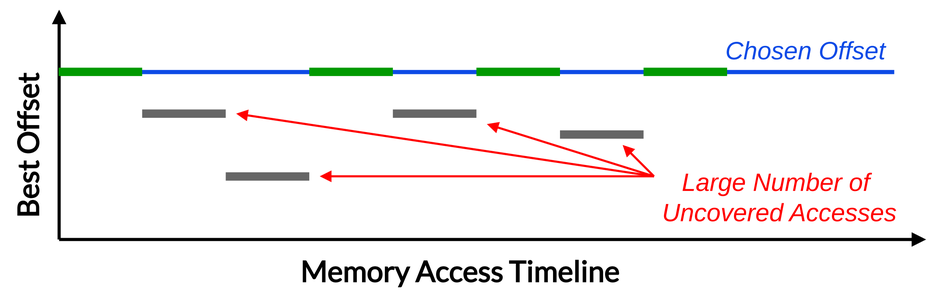


In SPECspeed 2017, **Caerus** outperforms **BOP** [1] (found in multiple modern processors) by 4.6% and **Berti** [2] (SoTA in literature) by 2.0% while maintaining low DRAM traffic.

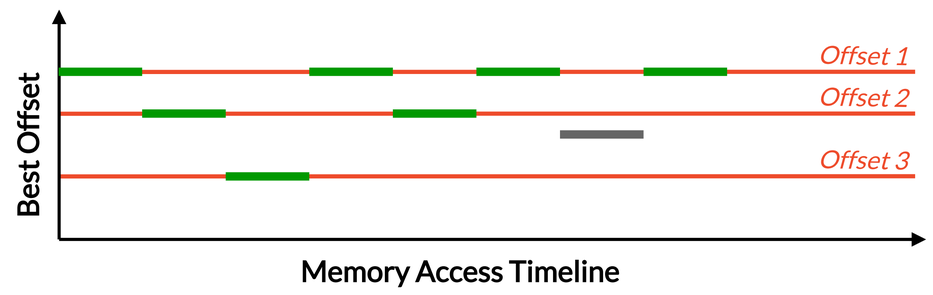
Artifact available @ github.com/jw38176/Caerus_Artifact

Contributions

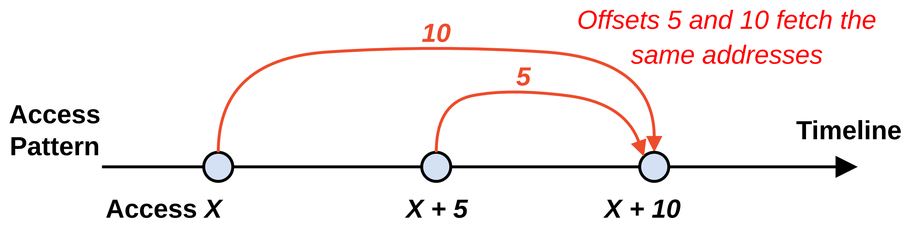
1. Problem: One offset is insufficient in capturing spatial relationships between cross-PC accesses



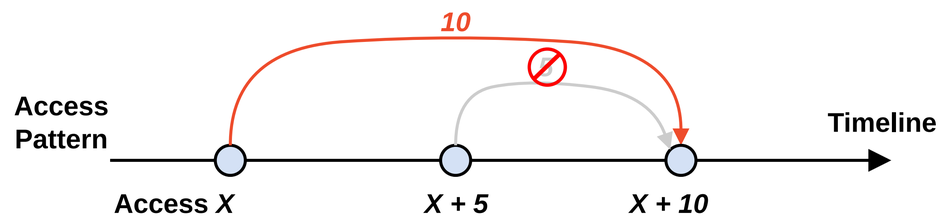
Solution: Extract **multiple global offsets** to maximise coverage



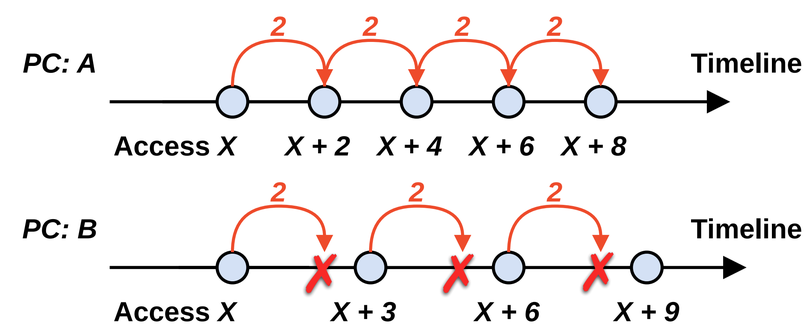
2. Problem: Multiple offsets might overlap



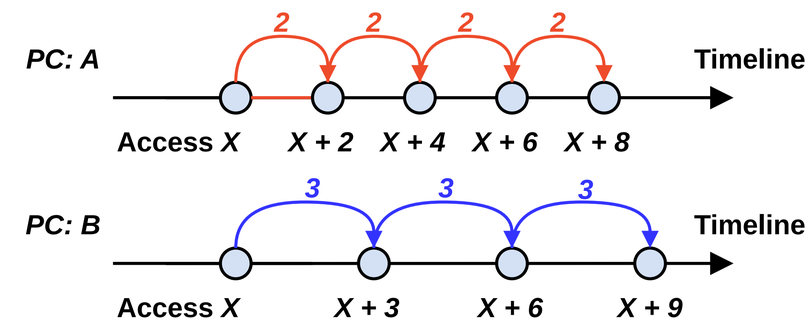
Solution: **Disregard redundant offsets** when training



3. Problem: High-coverage global offsets might be beneficial at one PC but not another



Solution: Perform **PC-local filtering** of offsets based on **per-PC accuracy**



[1] Pierre Michaud. Best-offset hardware prefetching. In 2016 IEEE International Symposium on High Performance Computer Architecture (HPCA), pages 469–480. IEEE, 2016.
[2] Agustín Navarro-Torres, Biswabandan Panda, Jesús Alastruey-Benedé, Pablo Ibáñez, Víctor Viñals-Yúfera, and Alberto Ros. Berti: an accurate local-delta data prefetcher. In 2022 55th IEEE/ACM International Symposium on Microarchitecture (MICRO), Pages 975–991. IEEE, 2022.

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